SMART STUDY PLANNER (CLI-Based AI Application)

Submitted by

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TITLE

Smart Study Planner using Rule-Based Intelligent Agent (CLI Application)

INTRODUCTION

Time management is a critical challenge for students, especially during examination periods. With multiple subjects, varying difficulty levels, and limited time, students often struggle to prioritize their studies effectively.

This project presents a Command Line Interface (CLI)-based Smart Study Planner that helps users organize and prioritize their study schedule. The system uses a rule-based intelligent approach to generate optimized study plans based on user inputs.

MOTIVATION OF THE PROJECT

As a student preparing for competitive exams, I personally experienced difficulty in managing time across multiple subjects. Deciding what to study first often led to confusion and inefficiency.

This motivated me to develop a system that can assist students in making better study decisions using simple AI techniques.

PROBLEM STATEMENT

Students lack an effective system to prioritize their study tasks based on urgency, difficulty, and workload, leading to inefficient preparation and increased stress.

OBJECTIVE OF THE PROJECT

- To develop a CLI-based study planner

- To implement a rule-based intelligent system
- To assist students in prioritizing subjects effectively
- To provide daily study recommendations

EXISTING METHODS

- Manual planning using notebooks
- To-do lists
- Calendar-based scheduling
- Mobile applications with fixed reminders

PROS AND CONS OF THE STATED METHODS

Pros:

- Easy to use
- No technical knowledge required
- Flexible

Cons:

- No intelligent prioritization
- Time-consuming
- Prone to human error
- Lack of adaptability

HARDWARE AND SOFTWARE REQUIREMENTS

Hardware:

- Basic computer or laptop

Software:

- Python (3.x)
- Command Prompt / Terminal
- Any text editor (VS Code recommended)

METHODOLOGY AND GOAL

The project follows a rule-based approach where subjects are prioritized based on:

- Difficulty level
- Days remaining until exam
- Estimated study hours

The goal is to simulate intelligent decision-making and provide an optimized study sequence.

FUNCTIONAL MODULES DESIGN AND ANALYSIS

1. Add Subject Module
   Allows user to input subject details
2. View Subject Module
   Displays all stored subjects
3. Edit Subject Module
   Enables modification of existing data
4. Delete Subject Module
   Removes unwanted subjects
5. Study Plan Generator
   Calculates priority and sorts subjects
6. Recommendation Module
   Suggests the most important subject for the day

ALGORITHM DEVELOPMENT

Step 1: Input subject details (name, difficulty, hours, exam date)
Step 2: Assign difficulty score
Step 3: Calculate urgency based on exam date
Step 4: Assign weight based on study hours
Step 5: Compute total priority score
Formula:
Priority = Difficulty Score + Urgency Score + Hours Weight
Step 6: Sort subjects in descending order
Step 7: Display study plan

SOFTWARE ARCHITECTURAL DIAGRAM

The system follows a modular structure:

- User interacts with CLI (main.py)
- Requests handled by planner.py
- Data stored in JSON file

Flow:
User → CLI → Logic Processing → Data Storage → Output

CODING

The project is implemented using Python. Key components include:

- File handling using JSON
- Functions for modular design
- Conditional logic for AI decision-making
- Sorting algorithms for prioritization

OUTPUT

Example Output:

Study Plan (High → Low Priority):

1. Physics (Priority: 6.5)
2. Mathematics (Priority: 5.2)
3. Chemistry (Priority: 4.8)

KEY IMPLEMENTATIONS OUTLINE OF THE SYSTEM

- Rule-based intelligent agent
- Priority scoring system
- CLI-based interaction
- Persistent data storage

- Modular code structure

SIGNIFICANT PROJECT OUTCOMES

- Improved understanding of AI concepts
- Practical implementation of intelligent systems
- Efficient study planning tool
- Enhanced programming and problem-solving skills

TESTING AND REFINEMENT

- Tested for invalid inputs
- Checked edge cases (empty data, incorrect values)
- Improved user prompts and error handling
- Refined priority calculation logic

PROJECT APPLICABILITY ON REAL-WORLD APPLICATIONS

- Student study planning
- Task prioritization systems
- Personal productivity tools
- Basic AI-based recommendation systems

CONTRIBUTION / FINDINGS OF THE PROJECT

- Demonstrated that simple AI techniques can solve real problems
- Highlighted importance of prioritization in productivity
- Showed effectiveness of rule-based systems

LIMITATION / CONSTRAINTS OF THE SYSTEM

- No graphical interface
- Depends on user input accuracy

- Uses basic AI (no machine learning model)

CONCLUSIONS

The Smart Study Planner successfully demonstrates how AI concepts like rule-based systems can be applied to real-world problems. It provides an efficient and practical solution for students to manage their studies effectively.

FUTURE ENHANCEMENTS

- Integration with calendar systems
- GUI-based interface
- Machine learning-based recommendations
- Mobile application version

REFERENCES

- Python Documentation (https://docs.python.org/)
- Course Materials (CSA2001)
- Basic AI Concepts – Intelligent Agents